Paleomagnetic Results from the Jurassic Topley Intrusions near Endako, British Columbia

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The Upper Jurassic Topley Intrusions intrude rocks of the tectonically stable Interior Plateau in central British Columbia. A stable primary remanent magnetization of both normal and reversed polarity was isolated after a.f. demagnetization in 19 of 22 sites (109 cores; 208 specimens) representing several of the plutonic units present in this sialic complex of quartz monzonite affinity near Endako, B.C. The unit mean remanence directions support some petrological correlations suggested by Carr, Bright, and White *et al.*, but do not support others. The pole position derived from 13 sites representing 6 plutonic units with a K–Ar radiometric age of 139 ± 4 m.y. (White *et al.*) is 128.6° E, 70.0° N (δ_m = 14.4°; δ_p = 11.4°). This position indicates a smooth polar wander path during the Jurassic between the well defined Upper Triassic and Lower Cretaceous pole positions. It also indicates that the northern Interior Plateau has not been tectonically rotated or translated since emplacement relative to the stable North American craton.

Les intrusions Topley (Jurassic supérieur), sont mises en place à travers les roches stables (tectoniquement) du Plateau intérieur en Colombie Britannique centrale. On a réussi a en mesurer un magnétisme rémanent primaire de polarité normale et inverse. Ceci a été réalisé après une démagnétisation (a.f.) dans 19 des 22 sites (109 carottes, 208 échantillons) qui représentent plusieurs des unités plutoniques présents dans ce complexe d'affinité monzonite quartzique près de Endako, C.B. La direction du magnétisme rémanent ainsi découvert, confirme les corrélations pétrologiques de Carr, Bright, et White *et al.* La position du pôle déduite des mesures sur 13 sites représentant 6 unités plutoniques (139 ± 4 m.a. daté au K–Ar) (White *et al.*) est 128.6° E, 70.0° N (δ_m = 14.4°; δ_p = 11.4°). Cette position indique un chemin de digression polaire lisse pendant le Jurassique entre le Trias supérieur d'une part et le Crétacé inférieur d'autre part, qui sont solidement établis. Ceci indique aussi que le Plateau intérieur n'a pas été pivoté tectoniquement et n'a pas souffert de mouvement de translation depuis l'emplacement relatif du Craton Nord Américain stable.
[Traduit par le journal]

Introduction

The Topley Intrusions outcrop in a NW–SE trending belt some 150 mi (240 km) long by 40 mi (64 km) wide, which is centered near the town of Endako in central British Columbia (Fig. 1). The name, Topley Intrusions, refers to a number of composite felsic batholiths and stocks that intrude tectonically stable rocks of the northern Interior Plateau. With few exceptions these intrusions show no petrographic evidence of secondary alteration or internal deformation. In the Endako area they have been geologically mapped and radiometrically dated in considerable detail. These features led to their selection for paleomagnetic study to determine whether the various intrusive phases could be classified by remanence direction as has been possible with similar rocks elsewhere (Symons 1972*b*). An additional object of this study was to add to the limited and inconsistent paleomagnetic data available for defining the magnetic polar wander path for the Jurassic period.

Geology

Armstrong (1949) mapped the geology around Endako as part of the Fort St. James map area. Based on limited paleontological and other geological evidence he concluded that the Topley Intrusions were emplaced after the Middle Permian, possibly before Upper Triassic, and probably before Lower Cretaceous. Tipper (1963*a*, *b*) found stratigraphic evidence southeast of Endako favoring a Lower Jurassic age of intrusion, consistent with the oldest (178 m.y.) of 5 K–Ar Topley age dates then available (Baadsgaard *et al.* 1961, Tipper 1963*b*). He attributed the 3 intermediate ages (163, 154, and 138 m.y.) to thermal metamorphic updating and the youngest age (63 m.y.) to a non-Topley Paleocene intrusion. Recently Tip-

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TABLE 1. Summary of geological and radiometric age data

Age	Unit	Lithology	Radiometric age[1] (m.y.)
Tertiary			
Eocene	Endako Group	volcanics: intermediate to basic dacite, basalt	
	Late Dikes[2]	porphyry	50 ± 2 51 ± 2
Paleocene	Tertiary stock**	granodiorite	63 ± 6[3]
Cretaceous			
Lower	Topley Intrusions		
	Youngest		
	Fraser*	quartz monzonite, granodiorite	112 ± 4, 114 ± 4
	Pond	quartz diorite	
Jurassic	Relatively Young		
Upper	Triangle[4]**	quartz monzonite	133 ± 5, 133 ± 5
	Stellako*	quartz monzonite	136 ± 5; 137 ± 5
	Intermediate		
	François*	granite, quartz monzonite	137 ± 5; 137 ± 6
	Casey*	quartz monzonite, alaskite, granite	139 ± 5; 137 ± 6; 139 ± 6[5]
	Relatively Old		
	Tatin[6]**	quartz monzonite	136 ± 5; 142 ± 6; 138 ± 11[3]
	Nithi[7]*	quartz monzonite	138 ± 5; 141 ± 6
	Glenannon*	quartz monzonite, granite, granodiorite	134 ± 5; 135 ± 6; 140 ± 6
	Endako ore body	hydrothermally altered quartz monzonite	140 ± 5[8]; 142 ± 5[8]; 142 ± 5[8]
	Endako*	quartz monzonite	140 ± 6; 141 ± 5; 143 ± 6; 141 ± 5[9]; 154 ± 13[3]
Middle	Oldest		
	Simon Bay*	quartz diorite, quartz monzonite, gabbro[10]	155 ± 6, 155 ± 6; 138 ± 6[11], 145 ± 8, 155 ± 8[12]; 163 ± 8
	Menard Stock[13]*	quartz latite porphyry	
Lower	Jurassic pluton**	diorite	178 ± 14[3]
Triassic	Takla Group	volcanics: acid tuffs, lavas, and breccias	

NOTES:
*Those units sampled in this study.
**Information derived from outside the area shown in Fig. 2.
[1]Radiometric ages in millions of years were determined by White *et al.* (1970) using the K–Ar method on biotite unless otherwise noted. Ages separated by a comma are replicate analyses on the same sample.
[2]Probably related to the Endako Group of Tertiary volcanic rocks (Kimura and Drummond 1969).
[3]Tipper (1963*b*).
[4]Probably northern equivalent of the Stellako phase (White *et al.* 1970).
[5]From dike equivalent of the Casey phase in the Endako ore body (White *et al.* 1968).
[6]Probably northern equivalent of the Glenannon phase (Carr 1965).
[7]Probably Nithi Mountain area equivalent of the Glenannon phase (Carr 1965).
[8]Secondary biotite from the alteration envelope of the Endako ore body dated (White *et al.* 1968).
[9]Primary biotite from the Endako orebody dated (White *et al.* 1968).
[10]Some primary foliation present in rocks of this unit.
[11]Plagioclase dated by K–Ar method.
[12]Hornblende dated by K–Ar method.
[13]Age in doubt; unit may be post Topley Intrusions (Kimura and Drummond 1969).

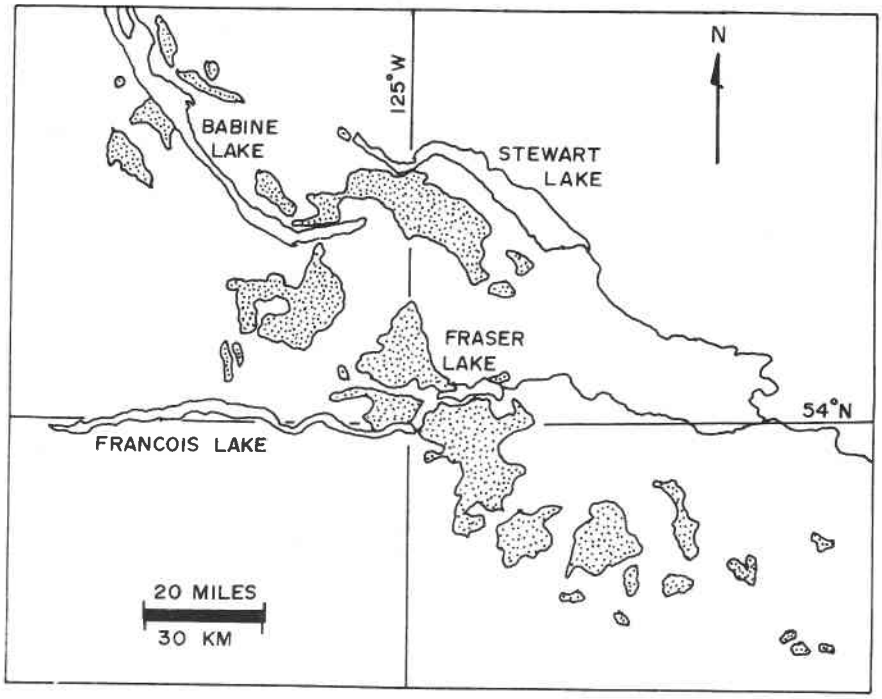


FIG. 1. Location of the Topley Intrusions.

per (1971) has found Topley Intrusions cutting the Lower Jurassic Tacheck group in the Smithers map area to the west which is in accord with the radiometric evidence of White *et al.* (1970).

Carr (1965) mapped the geology of a 250 mi² (640 km²) area around Endako at a scale of one mile to the inch. He concentrated on the Topley Intrusions which he divided into 13 map units based on petrological differences, internal contact features and geographic location. The geologic summary given in Table 1 follows that proposed by Carr (1965) except that some units falling outside the area of this study have been omitted, and that some modifications proposed by Bright (1967), Kimura and Drummond (1969), and White *et al.* (1970) have been included. Based on 25 K–Ar ages, White *et al.* (1970) concluded that the Topley Intrusions were emplaced either as one intermittent magmatic event or as three separate events, *i.e.*, the Simon Bay diorite complex was emplaced about 155 m.y. ago, the main potash-rich granitic and quartz monzonite units about 138 ± 3 m.y. ago, and the Fraser quartz monzonite about 114 m.y. ago. These rocks are coarse-grained, holocrystalline, and virtually unaltered in hand sample or thin section except for the slightly foliated Simon Bay diorite so that metamorphic updating of the K–Ar ages is unlikely. Cretaceous and Cenozoic extrusive and sedimentary rocks are gently dipping or flat-lying indicating little postintrusive structural deformation in this portion of the Pinchi geanticline (Armstrong 1949, Campbell and Tipper 1970). There are postintrusive block faults in the area (Fig. 2) as well as Tertiary volcanics and porphyry-copper type mineralization of the Endako molybdenum mine (Kimura and Drummond 1969) which indicate tensional tectonics. However, there is no geologic evidence for significant rotations which would affect the paleomagnetic remanence directions, and the paleomagnetic results within units across these faults similarly provides no clear evidence for significant rotations.

Method

Samples were collected from 22 sites representing most of the Topley intrusive units present in the Endako area (Fig. 2, Table 2). Five cores, each several meters apart and oriented

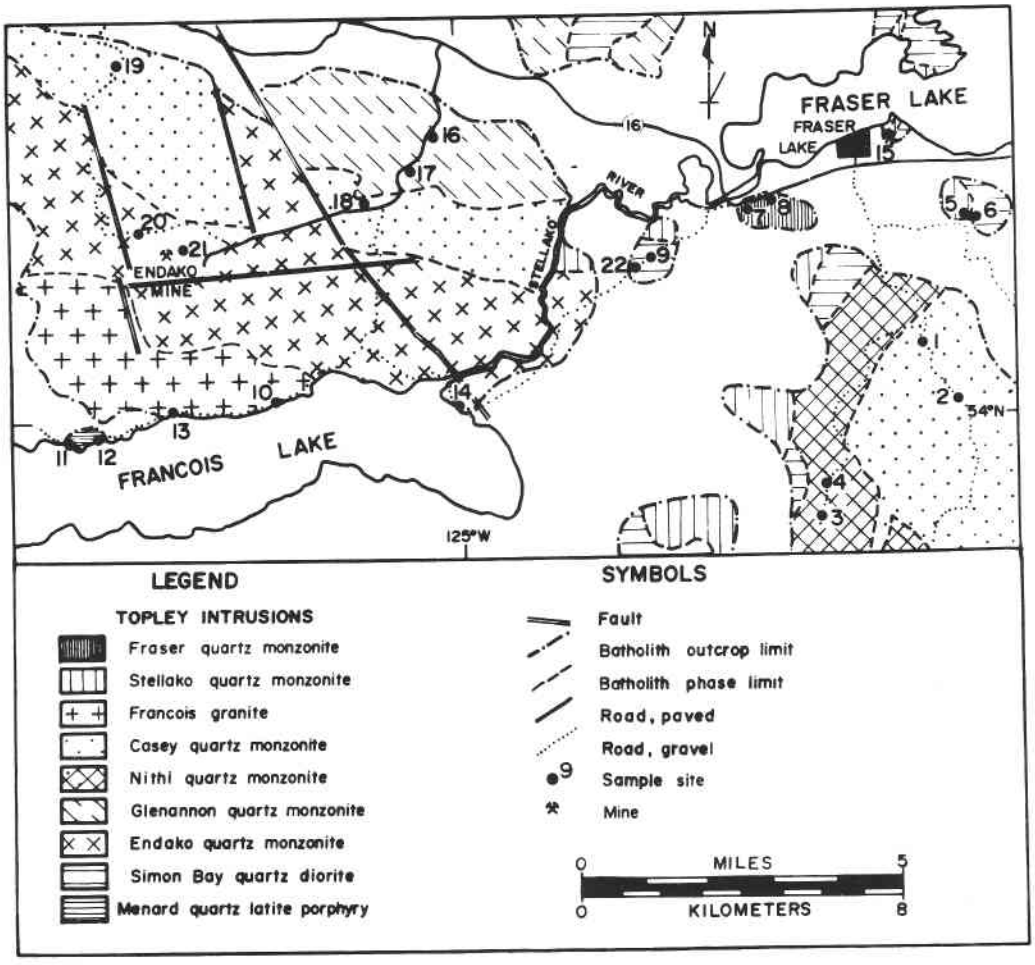


FIG. 2. General geology after Carr (1965) with modifications from Kimura and Drummond (1969) and sample site locations.

in situ by sun compass, were taken from each site except for one site with four cores. Two specimens were cut from each core, and the natural remanent magnetization (NRM) of each specimen was measured on a biastatic magnetometer (Larochelle and Christie 1967). One test specimen with representative NRM characteristics was selected from each site and demagnetized in an alternating field (a.f.) in steps of 50, 100, 200, 300, 400, 500, 650, and 800 Oe (peak) using apparatus described by Larochelle and Black (1965) with the remanence being remeasured after each step. Using the stability index (SI) method (Tarling and Symons 1967) and remanence intensity decay curves, an a.f. intensity was selected for each site in which its remaining specimens were demagnetized. Specimens from nine sites were demagnetized at two a.f. intensities. The remanent magnetization directions from the most successful cleaning treatment in terms of within-site grouping were used in subsequent analyses.

Twenty of the 109 collected cores were judged to yield unreliable remanence data and were rejected for the following reasons (Table 2). Two cores had weak remanent magnetization intensities ($< 5 \times 10^{-7}$ emu/cm^3) so that the magnetometer could give an erroneous direction. In 17 cores the angle θ between the remanence directions of the two specimens exceeded 20° which is an indication of inhomogenous within-core magnetization. One core gave an anomalous remanence direction deviating by more than three times the angular stan-

TABLE 2. Sampling and remanence data by sites

Site	Location °W	Location °N	Geologic unit	SI	*H*	Cores *N*	Cores *a*	Cores *b*	Cores *c*	Cores *n*	Site mean remanence *D*	*I*	*R*	*k*
1	124.82	54.02	Casey	2.3	100,(300)	5		2		3	357.2	76.2	2.965	57
2	124.81	54.00	Casey	5.0	200,(400)	5				5	335.2	73.7	4.973	149
3	124.86	53.98	Nithi	57.4	300	5				5	116.3	−62.9	4.986	287
4	124.86	53.99	Nithi	35.6	300	5				5	111.5	−57.2	4.989	361
5	124.80	54.04	Stellako	2.8	100,(500)	5				5	333.8	74.4	4.961	102
6	124.80	54.04	Simon Bay	3.2	200	5		3		2	294.4	48.4	1.672	3*
7	124.84	54.05	Fraser	4.4	100,(300)	5				5	315.5	78.1	4.996	987
8	124.88	54.05	Fraser	3.6	400	5				5	306.5	62.9	4.20	50
9	124.93	54.03	Simon Bay	14.3	300	5				5	299.3	46.8	4.912	45
10	125.07	54.00	François	2.1	200,(500)	5		2	1	2	343.8	33.1	1.992	119
11	125.17	54.00	Menard	5.4	(200),500	5		1		4	298.4	64.2	3.879	25
12	125.15	54.00	Menard	3.4	400	4	1			3	354.1	60.9	2.914	23
13	125.11	54.00	François	20.1	300	5				5	354.2	39.1	4.991	430
14	125.00	54.01	Endako	3.4	200	5				5	331.3	49.5	4.970	133
15	124.83	54.06	Stellako	9.9	100,(400)	5				5	332.0	73.7	4.981	205
16	125.01	54.06	Glenannon	1.7	200	5		5		0				
17	125.02	54.06	Glenannon	1.1	100,(300)	5		1		4	324.0	79.8	3.976	123
18	125.04	54.05	Casey	15.4	200	5				5	334.6	58.6	4.988	343
19	125.13	54.08	Casey	4.1	(100),400	5	1	1		3	330.9	85.2	2.973	74
20	125.12	54.04	Endako	5.0	300	5		2		3	151.0	0.9	2.528	4*
21	125.10	54.04	Endako	24.5	100	5				5	158.8	−41.8	4.997	1212
22	124.93	54.03	Simon Bay	14.1	100	5				5	300.7	48.6	4.743	16

NOTES:
°W, °N = Longitude west and latitude north of site in degrees.
SI = Stability index.
H = A.F. demagnetization peak field used (not used) in oersteds.
Cores = Collected (*N*); rejected because (a) of weak remanence ($< 5 \times 10^{-7}$ emu) or (b) of inhomogenous within-core remanence direction ($\theta > 20°$), or (c) of anomalous remanence direction for site; and, used for site mean remanence statistics (*n*).
D, *I*, *k*, *R* = Declination, inclination, and precision parameter of site mean remanence resultant vector giving *n* cores unit weight.
*Site remanence inhomogenous with $k < 15$.

dard deviation from the mean direction of the remaining cores at the site. This rejection process led to the total elimination of only one site (16). As a final step, the two sites (6 and 20) with a precision parameter (k) for their mean remanence direction of less than 15 were considered inhomogenously magnetized and rejected. Thus a characteristic site mean remanence direction is thought to have been isolated for 19 of the original 22 sites.

Discussion

There are several points of evidence that suggest that the isolated remanence in the valid sites is stable and primary: (1) the rocks show no evidence of metamorphism except for the slightly foliated Simon Bay diorite either in hand sample or thin section (White *et al.* 1970) during which a secondary remanent magnetization could be introduced, (2) the isothermal remanence components (IRM) removed during a.f. demagnetization are uniformly aligned parallel to the present earth's magnetic field—the remanence intensity drops quickly to a mean of 42.8% of the original NRM intensity by 100 Oe as the IRM components are removed and then drops slowly to 25.1% by 300 Oe as the stable characteristic remanence components are isolated, (3) the average SI value (11.5) and average precision parameter ($k = 165$) determined for the sites indicates a stable homogenous magnetization, (4) the coherent within-unit directions which diverge from each other and from the present earth's magnetic field direction militate against the presence of secondary remanence, (5) the antiparallel normal and reversed directions found in the Endako unit and the reversed directions in the Nithi unit virtually demand a stable primary remanence, and (6) as will be shown, the location of the derived Upper Jurassic pole position.

The number of valid site mean remanent magnetization directions available for each of the Topley units is not always sufficient for rigorous inter-unit comparison by variance

TABLE 3. Remanence data by units

Unit(s)	Unit mean remanence						Pole position	
	n'	R	D'	I'	k	α_{95}	Long °E	Lat. °N
Menard	2	1.9523	327.9	65.3	21	21.7	140.0	68.8
Simon Bay	2	1.9997	300.0	47.7	3182	1.8	139.7	40.3
Endako	2*	1.9934	335.3	45.7	152	8.2	98.8	57.6
Nithi	2+	1.9971	293.7	60.1	344	5.4	156.6	45.1
Glenannon	1		324.0	78.9			200.0	68.0
Casey	4	3.9402	339.2	73.7	50	9.9	179.3	77.4
François	2	1.9919	348.8	36.2	123	9.0	73.4	55.0
Stellako	2	1.9999	332.8	74.1	17222	0.8	179.1	74.2
Fraser	2	1.9819	309.3	70.6	55	13.4	167.5	61.1
Upper Jurassic**	13†	12.3384	332.5	63.3	18	9.1	128.6	70.0

NOTE:
The following heading abbreviations are used: Number of sites (n'); length of resultant vector (R); declination (D'), inclination (I'), precision parameter (k) and radius of cone of 95% confidence (α_{95}) of unit mean remanent magnetization in degrees; longitude east (Long. °E), latitude north (Lat. °N) of pole position in degrees.
*Directions of one site reversed through 180° to the normal position before calculation of data.
**Sites from Endako, Nithi, Glenannon, Casey, François, and Stellako phases included.
+Directions of both sites reversed through 180° to the normal position before calculation of data.
†Directions of three sites reversed through 180° to the normal position before calculation of data.

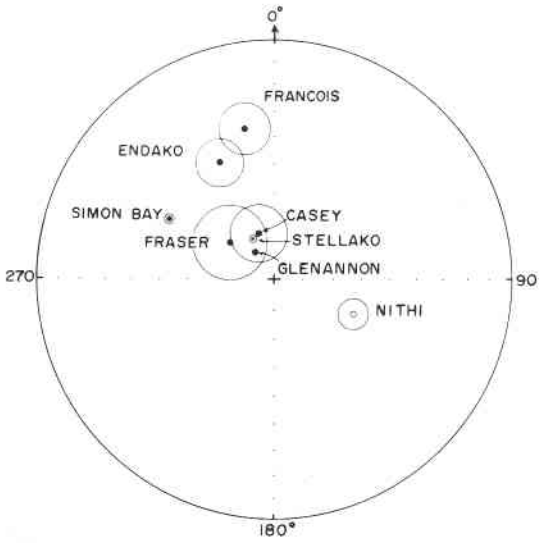


FIG. 3. Unit mean remanent magnetization directions for the Topley Intrusions surrounded by their cone of 95% confidence. Polarity of the direction is given by a solid circle for normal (downward) directions and an open circle for reversed (upward) directions as listed in Table 3. The present earth's magnetic field direction is under the S in Stellako.

ratio statistics. However, the paleomagnetic method with additional sampling does appear to have potential for distinguishing between the units. For example, the unit mean remanence directions (Fig. 3, Table 3) for several units such as the Simon Bay and Endako indicate distinctive remanence populations. It is doubtful that the Nithi and Glenannon are contemporaneous (Carr 1965, Bright 1967, White *et al.* 1970) because the former reflects a reversed earth's magnetic field which is not even antiparallel to the latter's normal field position. Also, the Casey and Francois give divergent unit mean remanence directions indicating that simply grouping them together on the basis of similar petrology as "relatively young intrusions" (White *et al.* 1970) may well be erroneous. Conversely, it does appear valid to correlate the Casey unit from Nithi Mountain complex on the east of the sampling area (Fig. 2) across the Stellako River to the Endako complex on the west side as both areas of the unit give very similar directions. The directions also favor grouping the François granite with the Endako unit as Bright (1967) suggests rather than with the Casey unit as White *et al.* (1970) suggest. The large α_{95} limits found for the Menard stock do not allow for speculation as to its age relative to the Topley Intrusions.

Using the radiometric evidence summarized in Table 1 as a guide, it is reasonable to derive a valid Upper Jurassic (139 ± 4 m.y.) pole position. Excluding sites in the Menard unit of uncertain age, the older Simon Bay unit, and the younger Fraser unit, the remaining 13 valid sites come from units (Endako, Nithi, Glenannon, Casey, François, and Stellako) with radiometric ages ranging from 135 to 143 m.y. Giving each site mean remanence direction unit weight and rotating reversed directions through

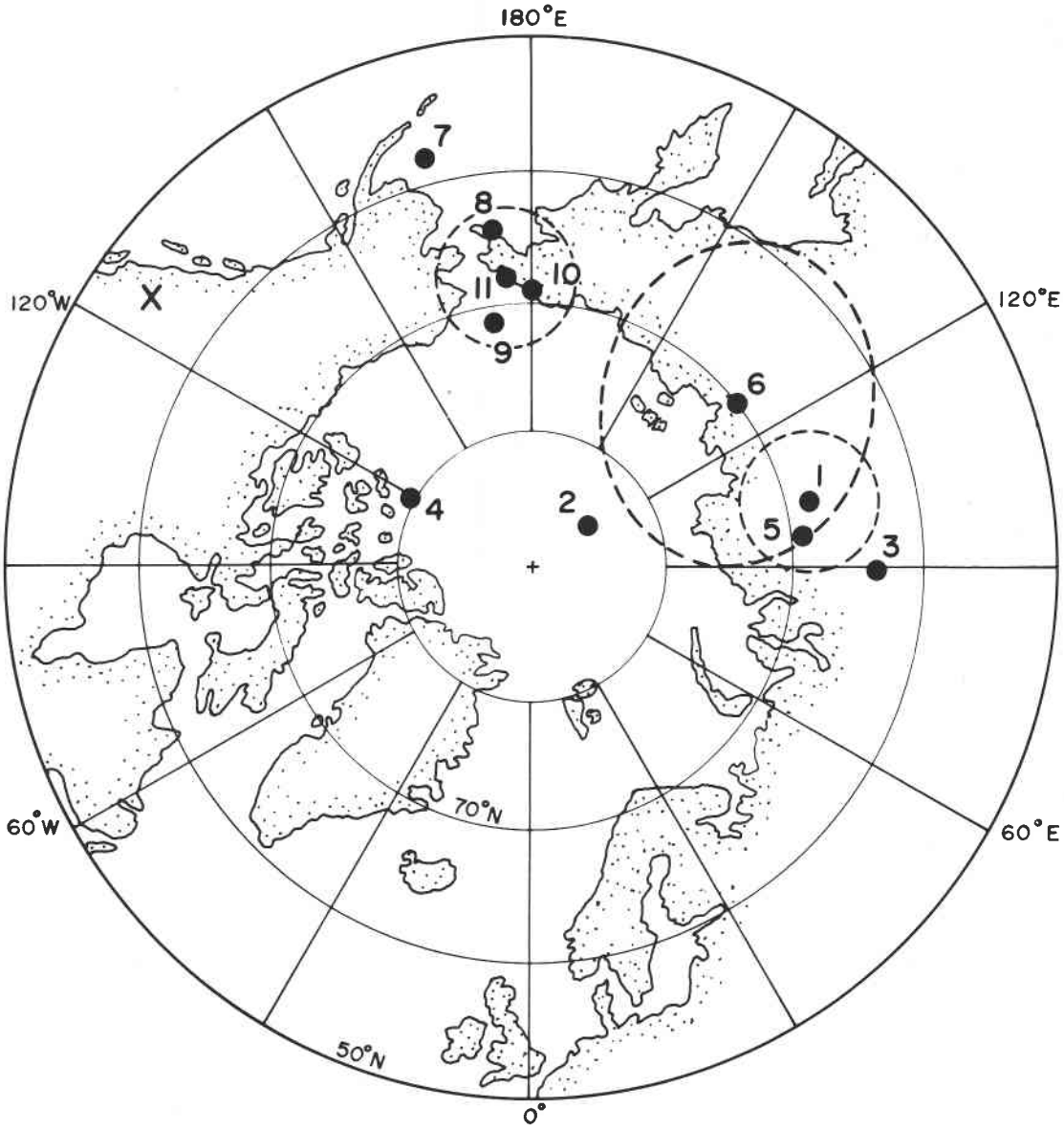


FIG. 4. Polar projection of the northern hemisphere north of latitude 50° N showing the pole positions listed in Table 4. The oval of 95% confidence is shown for the Upper Triassic mean pole (1), the Topley Intrusions pole (6) and the Lower Cretaceous mean pole (11). The sampling site is an X.

180° to the normal position, the resulting mean Upper Jurassic direction is 332.5°, 63.3 (k = 18; α_{95} = 9.1°; Table 3). Using the dipole formula, the resulting pole position is 128.6° E, 70.0° N (δ_m = 14.4°; δ_p = 11.4°). As shown in Fig. 4 and Table 4, this position is intermediate between the Upper Triassic and Lower Cretaceous clusters of reliable pole positions from North American formations. For this reason and because there is no geologic evidence for significant tectonic rotation within the Topley Intrusions or within the tectonically stable region of the Interior Plateau which they intrude, this position is regarded as a reliable Upper Jurassic pole. As such it provides additional evidence in support of the hypothesis of clock-

TABLE 4. Upper Triassic to Lower Cretaceous pole positions for North American rock formations

Pole	Formation unit	Age	Pole position				Reference
			Long. °E	Lat. °N	δ_m	δ_p	
1	Upper Triassic mean	uTr, 190 m.y.	103	68	6	6	See notes
2	White Mountain volcanics	1J, 180 m.y.	126	85	6	5	Opdyke and Wensink (1966)
3	Kayenta red beds	uTr–1J	88	63			Steiner and Helsley (1971)
4	Island Intrusions granodiorite	mJ–uJ, 159 ± 10 m.y.	240	79	11	10	Symons (1971*a*)
5	Summerville red beds	uJ	97	69			Steiner and Helsley (1971)
6	Topley Intrusions monzonite	uJ, 139 ± 4 m.y.	129	70	14	11	this paper
7	Bucks Batholith granodiorite	uJ, 138 ± 6 m.y.	195	58	8	8	Grommé *et al.* (1967)
8	Mt. Ascutney gabbro	1K, 130 ± 5 m.y.	187	64			Opdyke and Wensink (1966)
9	Monteregian Hills carbonatite	1K, 124 ± 3 m.y.	189	71	4	3	Larochelle (1969)
10	Isachsen diabase	1K, 106 ± 4 m.y.	180	69	14	4	Larochelle and Black (1963); Larochelle *et al.* (1965)
11	Lower Cretaceous mean	1K	185	68	6	6	poles 8, 9, and 10

NOTES:
The abbreviations used are: Triassic (Tr), Jurassic (J), Cretaceous (K), Lower (l), Middle (m), Upper (u), and million years (m.y.) for radiometric age dates. The pole position is given in degrees of longitude east, latitude north, and the semi-axes (δ_m, δ_p) of the oval 95% confidence. The 11 poles are listed in Symons (1971*a*) and include the Chinle formation, North Mountain basalt, Grand Manan lavas, Great Dyke of Nova Scotia, Meridan Formation, the Newark Group of Connecticut, New Jersey, and Pennsylvania, the Mistastin Lake volcanics, and the Appalachian diabase dikes.

wise rotation of the southern end of the Interior Plateau proposed by Symons (1971*b*, 1972*a*) from his study of the paleomagnetism of the Triassic Guichon batholith. If the western Cordillera has been translated northward by some 25° to 50° relative to the North American craton as Beck and Noson (1972) and Irving and Yole (1972) have suggested, then it is apparent that the Intermontane Belt which contains the Interior Plateau has not participated in this movement.

Comparison of the Topley pole (pole 6) (Table 4, Fig. 4) with other Jurassic pole positions from North American rocks is difficult because of lack of reliable data. The pole from the Appalachian diabase dikes which de Boer (1967) suggests may be Jurassic because of its position, has been assigned to the Upper Triassic (pole 1) in accordance with geological correlation and radiometric evidence (McElhinny 1968). Similarly the pole from the Mistastin Lake volcanics which Currie and Larochelle (1969) suggest may be Lower Jurassic because of its proximity to the White Mountain volcanics pole (pole 2) (Opdyke and Wensink 1966), has been assigned to the Triassic on the basis of its 202 ± 13 m.y. K–Ar age. Steiner and Helsley (1971) have: (1) revised the Upper Triassic – Lower Jurassic Kayenta red beds pole position (pole 3) reported by Collinson and Runcorn (1960) who did not use any demagnetizing treatment to remove unstable components, (2) found that the Middle or Upper Jurassic Carmel red beds pole position reported by Collinson and Runcorn (1960) was derived from paleomagnetically unstable rocks, and (3) found a reliable pole position (pole 5) for the Upper Jurassic Summerville red beds. Unfortunately, the confidence limits for the Kayenta and Summerville red beds are not reported by Steiner and Helsley (1971), and the former predates intrusion of the Topley Intrusions by some 40 or 50 m.y. Similarly, the reliable White Mountain volcanics pole predates the Topley by some 40 m.y. The pole position for the Middle to Upper Jurassic Island Intrusions (pole 4; Symons 1971*a*) may be reliable or may reflect minor clockwise rotation. The pole position of the Upper Jurassic Guadelupe Mountain igneous complex (Grommé *et al.* 1967) has not been included because of its sparse sampling (4 sites) and

large oval of 95% confidence ($\delta_m = 27°$; $\delta_p = 27°$). In the same vein while the Upper Jurassic – Lower Cretaceous Bucks Batholith pole position (pole 7) is reasonably consistent with other reliable Lower Cretaceous pole positions (8–11), it does come from the western Sierra Nevada, which has been a tectonically active area since emplacement of the batholith. Finally the Franciscan Formation poles (Grommé and Gluskoter 1965, Saad 1969) deviate markedly in directional consistency for reasons not yet fully understood, and may in fact be Late Cretaceous in age. Considering this array of Jurassic data, it is concluded that the Topley Intrusions (pole 6) provide as reliable an Upper Jurassic pole position as any available. Its position midway between the Upper Triassic mean pole (1) and the Lower Cretaceous cluster (8–10) of poles from cratonic North America indicates a nearly linear movement of the Earth's magnetic field pole from the Upper Triassic to Lower Cretaceous during the Jurassic. This is contrary to the static Triassic–Jurassic pole position interpretation of Steiner and Helsley (1971), and to the "hairpin turn" interpretation of Irving and Park (1972).

One final point is apparent. It is essential that all phases of a sialic complex be sampled on a broad geographic basis if a reliable paleomagnetic pole position is to be determined. Individually the Upper Jurassic phases or units would probably have given a biased and inaccurate position.

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